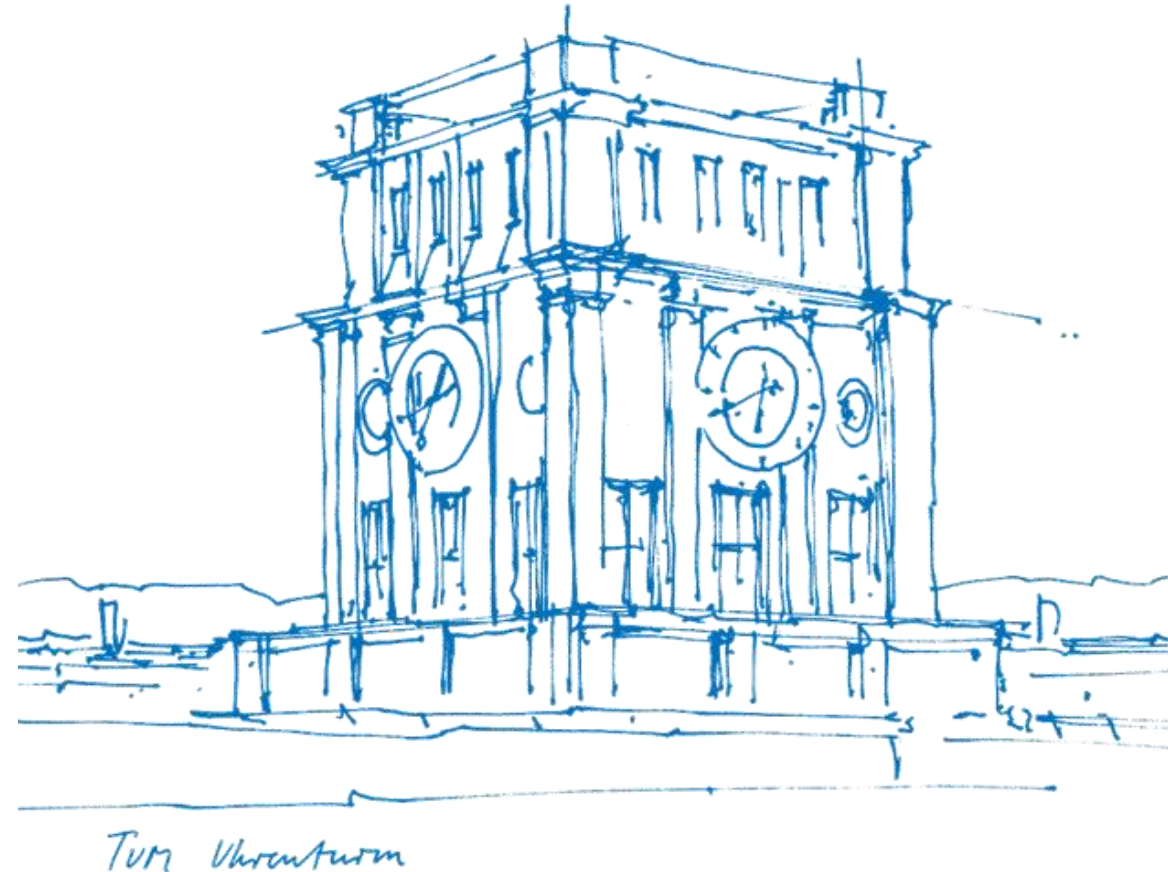


Tutorial 12

Hypothesis Testing

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About Tutorial

- **Tutorial:** In-Person, 2h by two time periods per week
 - 12:00-14:00 Tue, 03.11.018 (Group B33)
 - 10:00-12:00 Fri, 01.06.011 (Group E24)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Materials of this tutorial:** Please scan the QR-Code 

- **PART I.** Recap of the Lecture
- **PART II.** Exercises from the Exercise Sheet
- **PART III.** Additional Exercise (with solution)



I. Recap

• Hypothesis Testing

- ① - Null-Hypothesis (H_0) and Alternative-Hypothesis (H_1)
- ② - Type I and Type II Error
- ③ - One-sided and Two-sided Test
- ④ - z-Test and t-Test
- ⑤ - Significance Level and Rejection Region
- ⑥ - Power of a Test
- ⑦ - Neyman-Pearson Lemma (Likelihood ratio test, lrt)

true state	decision	
	fail to reject H_0	reject H_0
H_0 is true	correct decision	type I error
H_1 is true	type II error	correct decision

Fig 1. Error Types

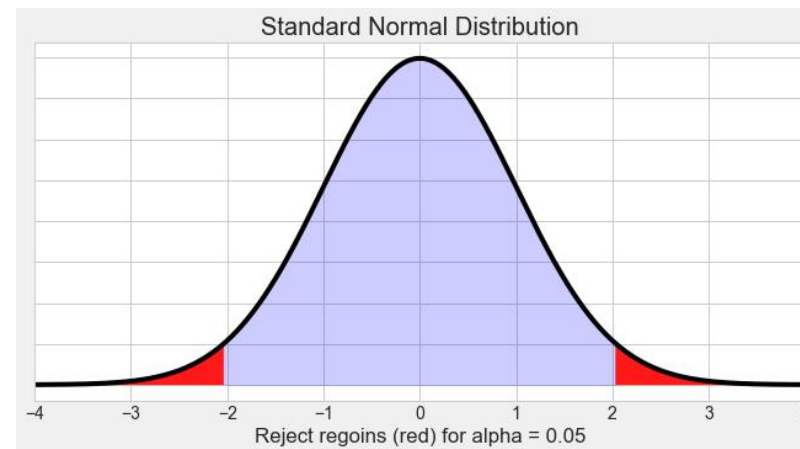


Fig 2. Rejection Region

$\Phi(x)$	0,00	0,01	0,02	0,03	0,04	0,05	0,06	0,07	0,08	0,09
0,0	0,500	0,504	0,508	0,512	0,516	0,520	0,524	0,528	0,532	0,536
0,1	0,540	0,544	0,548	0,552	0,556	0,560	0,564	0,567	0,571	0,575
0,2	0,579	0,583	0,587	0,591	0,595	0,599	0,603	0,606	0,610	0,614
0,3	0,618	0,622	0,626	0,629	0,633	0,637	0,641	0,644	0,648	0,652
0,4	0,655	0,659	0,663	0,666	0,670	0,674	0,677	0,681	0,684	0,688
0,5	0,691	0,695	0,698	0,702	0,705	0,709	0,712	0,716	0,719	0,722
0,6	0,726	0,729	0,732	0,736	0,739	0,742	0,745	0,749	0,752	0,755
0,7	0,758	0,761	0,764	0,767	0,770	0,773	0,776	0,779	0,782	0,785
0,8	0,788	0,791	0,794	0,797	0,800	0,802	0,805	0,808	0,811	0,813
0,9	0,816	0,819	0,821	0,824	0,826	0,829	0,831	0,834	0,836	0,839
1,0	0,841	0,844	0,846	0,848	0,851	0,853	0,855	0,858	0,860	0,862
1,1	0,864	0,867	0,869	0,871	0,873	0,875	0,877	0,879	0,881	0,883
1,2	0,885	0,887	0,889	0,891	0,893	0,894	0,896	0,898	0,900	0,901
1,3	0,903	0,905	0,907	0,908	0,910	0,911	0,913	0,915	0,916	0,918
1,4	0,919	0,921	0,922	0,924	0,925	0,926	0,928	0,929	0,931	0,932
1,5	0,933	0,934	0,936	0,937	0,938	0,939	0,941	0,942	0,943	0,944
1,6	0,945	0,946	0,947	0,948	0,949	0,951	0,952	0,953	0,954	0,954
1,7	0,955	0,956	0,957	0,958	0,959	0,960	0,961	0,962	0,962	0,963
1,8	0,964	0,965	0,966	0,966	0,967	0,968	0,969	0,969	0,970	0,971
1,9	0,971	0,972	0,973	0,973	0,974	0,974	0,975	0,976	0,976	0,977
2,0	0,977	0,978	0,978	0,979	0,979	0,980	0,980	0,981	0,981	0,982
2,1	0,982	0,983	0,983	0,983	0,984	0,984	0,985	0,985	0,985	0,986
2,2	0,986	0,986	0,987	0,987	0,987	0,988	0,988	0,988	0,989	0,989
2,3	0,989	0,990	0,990	0,990	0,990	0,991	0,991	0,991	0,991	0,992
2,4	0,992	0,992	0,992	0,992	0,993	0,993	0,993	0,993	0,993	0,994
2,5	0,994	0,994	0,994	0,994	0,994	0,995	0,995	0,995	0,995	0,995
2,6	0,995	0,995	0,996	0,996	0,996	0,996	0,996	0,996	0,996	0,996
2,7	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997	0,997
2,8	0,997	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,998
2,9	0,998	0,998	0,998	0,998	0,998	0,998	0,998	0,999	0,999	0,999

Fig 3. Standard Normal Table

II. Exercise Sheet

Exercise 1 (hypothesis test for a mean). A factory produces light bulbs, and the historical average lifetime is $\mu_0 = 1200$ hours with a known standard deviation of $\sigma = 150$ hours. A new manufacturing process is introduced, and the manager wants to test if the average lifetime has changed. A sample of $n = 36$ light bulbs produced by the new process has a sample mean lifetime of $\bar{x}_n = 1250$ hours. Assume the lifetimes are normally distributed.

- (i) State the null hypothesis H_0 and the alternative hypothesis H_1 for this test.
- (ii) Calculate the value of the appropriate test statistic.
- (iii) Would you reject the null at a significance level of $\alpha = 0.05$? What about at $\alpha = 0.001$? What does that mean for the confidence you should have in the rejection?

Hint: Follow to the normal distribution table, $z_{0.025} \approx 1.96$, $z_{0.0005} \approx 3.29$

II. Exercise Sheet



Exercise 2 (Neyman-Pearson test for exponential distribution). Suppose we observe a single sample X from an exponential distribution $\text{Exp}(\lambda)$. We want to test the simple null hypothesis $H_0 : \{\lambda_0\}$ against the simple alternative hypothesis $H_1 : \{\lambda_1\}$, where $\lambda_1 > \lambda_0 > 0$.

- (i) Write down the likelihood ratio $\Lambda(x) = \frac{L(\lambda_1|x)}{L(\lambda_0|x)}$.
- (ii) According to the Neyman-Pearson lemma, the most powerful test of size α rejects H_0 if $\Lambda(x) > k'$ for some constant k' . Show that this rejection region is equivalent to $x < c$ or $x > c$ for some constant c . Determine which form is correct for $\lambda_1 > \lambda_0$.
- (iii) For a significance level α , determine the explicit form of the critical value c for the rejection region.
- (iv) What is the power of this test?

II. Exercise Sheet



Exercise 3 (one-sample t-test for a mean). A brewery claims that its bottles contain exactly 330 ml of beer. A skeptical student measures the contents of $n = 10$ randomly selected bottles and finds a sample mean of $\bar{x}_{10} = 328.5$ ml and a sample variance of $s_{10}^2 = 4$ ml². Assuming the amount of beer in a bottle is normally distributed.

- (i) State the null and alternative hypotheses for testing if the true mean amount of beer is different from 330 ml.
- (ii) What is the appropriate test for this situation? State the test statistic and its distribution under the null hypothesis.
- (iii) Perform the test at a significance level of $\alpha = 0.05$. Should the student reject the brewery's claim? [Hint: You may use that the 0.975-quantile of a t -distribution with 9 degrees of freedom is $t_{9,0.025} \approx 2.262$.]

II. Exercise Sheet



Exercise 4 (type II error and power of a z-test). Suppose you are testing the mean of a normal population $\mathcal{N}(\mu, \sigma^2)$ with known variance $\sigma^2 = 9$. You collect $n = 25$ samples. You want to test the null hypothesis $H_0 : \mu \leq 10$ against the alternative $H_1 : \mu > 10$ at a significance level of $\alpha = 0.01$.

- (i) Determine the rejection region for this test. You may use that the 0.99-quantile of a standard normal distribution is $z_{0.01} \approx 2.33$.
- (ii) Calculate the probability of a type II error, β , if the true mean is actually $\mu_a = 11.5$.
- (iii) What is the power of the test against the alternative $\mu_a = 11.5$?

III. Additional Exercise

Exercise 5. A beverage manufacturer wants to determine whether their new Spezi recipe is better received by customers than their old recipe. According to previous surveys, the taste of the old recipe scored $\mu_0 = 7$ out of 10 taste points. A new survey is commissioned where 100 participants rate the taste of the new recipe on a scale from 1 to 10. You may assume that each person's rating is independent and identically distributed with unknown expected value μ . Using an appropriate hypothesis test, determine whether the claim $\mu \leq \mu_0$ can be rejected at a 1% significance level, given the following survey results:

Rating	1	2	3	4	5	6	7	8	9	10
Count	3	0	1	2	4	11	17	20	24	18

Note: You may approximate $t_{k,\alpha}$ for $k \geq 30$ with z_α , where z_α is the α -quantile of the standard normal distribution.

III. Additional Exercise

Let $X_1, \dots, X_{100}$ be the independent and identically distributed ratings from the 100 participants. As given, $\mathbb{E}[X_i] = \mu$. We perform a t -test since the variance of X_i is unknown and $n = 100 \geq 30$ is sufficiently large. The null hypothesis is $H_0 : \mu \leq \mu_0$. Our test statistic T is:

$$T = \frac{\bar{X} - 7}{S} \cdot 10,$$

where $\bar{X}$ is the sample mean and S is the sample standard deviation. Using the table values, we calculate:

$$\bar{X} = \frac{3 \cdot 1 + 0 \cdot 2 + 1 \cdot 3 + 2 \cdot 4 + 4 \cdot 5 + 11 \cdot 6 + 17 \cdot 7 + 20 \cdot 8 + 24 \cdot 9 + 18 \cdot 10}{100} = 7.75$$

and

$$S = \sqrt{\frac{3 \cdot (1 - 7.75)^2 + 0 \cdot (2 - 7.75)^2 + \dots + 18 \cdot (10 - 7.75)^2}{99}} = \sqrt{\frac{390.75}{99}} \approx 1.99.$$

The test statistic T is therefore:

$$T \approx \frac{0.75}{1.99} \cdot 10 \approx 3.77.$$

According to the lecture, the null hypothesis can be rejected at a 0.01 significance level if T is greater than $t_{99,0.99}$. As per the note, we can approximate $t_{99,0.99}$ with $z_{0.99}$. From the table in the appendix, we find that $z_{0.99} \approx 2.31$. We can therefore reject the null hypothesis at the 0.01 significance level.